

Express View on DeepSeek: Small and smart

If artificial intelligence is the new battlefield, when was the first shot fired? Was it last week, when a Chinese AI lab released a Large Language Model (LLM), developed at a **fraction** of the cost of the ones built by Silicon Valley, and yet capable of beating these same models on several **benchmarks**? Or was it this week, when Wall Street opened to a **bloodbath**, with US stocks losing a reported \$1 trillion in value, including chip-maker Nvidia's \$600-billion drop? **Perhaps** it can be **traced way back to** October 2022, when the US, in an effort to retain its **edge**, banned the export to China of advanced chips used for AI development? As questions **pile up** on the **implications** of China's DeepSeek R1 and the future of AI, one thing is clear: America's **lead** in the race **has shrunk considerably**. It seems to have happened, **moreover**, in **the blink of an eye**, catching the US and Silicon Valley — as **indeed**, the world — completely by surprise.

Yet, it is not as if DeepSeek, with a logo **featuring** a blue whale, **surfaced out of nowhere**. A **minnow** compared to the Big Four in China's tech sector (Baidu, Alibaba, Tencent and Xiaomi), DeepSeek made **ripples** last year when it released its V3 model, which was described by Andrej Karpathy, one of the founders of OpenAI, as a "frontier-grade LLM, trained on a joke of a budget" (a "**mere**" \$6 million, compared to the over \$500 million that OpenAI has reportedly spent on ChatGPT). The January 20 **debut** of R1, a reasoning model developed on a **shoestring** budget and with second-rate chips which **can**, **nonetheless**, perform as well as OpenAI's most advanced model, **shook** one of the central **tenets** of AI development in the US — that it takes billions of dollars to produce **cutting-edge** technology.

That lack of access to first-rate chips couldn't **hinder** DeepSeek also **shows** the limits of America's "**small yard, high fence**" **approach**, and that smaller, **nimbler** competitors could easily get within **heel-snapping** distance. The open-source **strategy** adopted by Chinese labs, including DeepSeek, **as opposed to** the **proprietary** model favoured by those in the US (except, notably, Meta) also **hints at** another way in which the rules of the game could be rewritten by China, as more global talent — and, **indeed**, other countries **nursing** their own AI **ambitions** — **gravitate** towards free-to-use models.

India, which talks of building "**sovereign AI**", **may** find hope in the DeepSeek story, that it is possible to do more with less. Yet, even as it **puts its weight** being the ambition and talent of the homegrown tech sector, and puts in the Rs 10,300 crore **committed** under the IndiaAI mission in March 2024, the government must also find a way to ensure access to the best AI hardware. If the future will be defined by AI, India must be a part of it.

[Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

Vocabulary

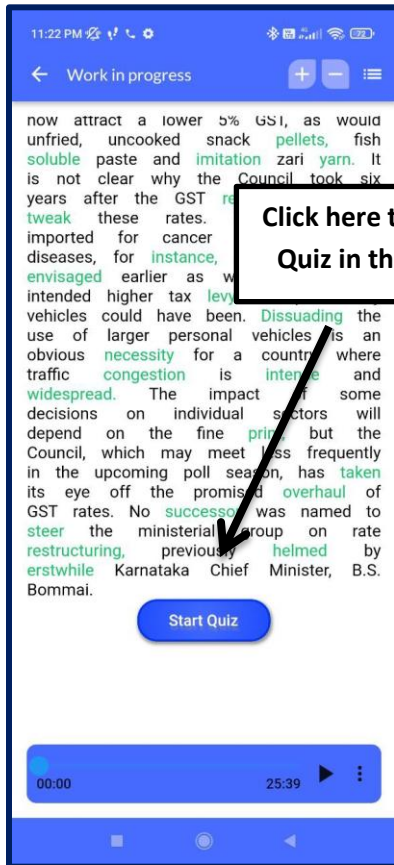
1. **Fraction** (noun) – part, portion, segment, piece, snippet छोटा हिस्सा
2. **Benchmark** (noun) – standard, yardstick, criterion, reference point, measure मानक
3. **Bloodbath** (noun) – meltdown, massacre, carnage, catastrophic downfall, brutal crash विनाशकारी गिरावट / व्यापक तबाही
4. **Perhaps** (adverb) – maybe, possibly, conceivably, it could be, potentially शायद
5. **Trace way back to** (phrase) – date back to, originate in, find origins in, track down to, go back to की शुरुआत तक पीछे जाना
6. **Edge** (noun) – advantage, lead, upper hand, vantage, position of strength बढ़त
7. **Pile up** (phrasal verb) – accumulate, gather, collect, amass, stack up जमा होना
8. **Implication** (noun) – consequence, outcome, effect, result, ramification परिणाम
9. **Shrink** (verb) – reduce, diminish, lessen, contract, decrease छोटा होना
10. **Considerably** (adverb) – significantly, substantially, markedly, notably, greatly काफी हद तक
11. **Moreover** (adverb) – furthermore, in addition, also, besides, plus इसके अलावा
12. **The blink of an eye** (phrase) – instantaneously, in a flash, in no time, very quickly, in a split second पलक झपकते ही
13. **Indeed** (adverb) – truly, certainly, in fact, surely, unquestionably वास्तव में
14. **Feature** (verb) – highlight, present, showcase, exhibit, include प्रदर्शित करना
15. **Surface** (verb) – emerge, come to light, appear, manifest, come up सामने आना
16. **Out of nowhere** (phrase) – unexpectedly, suddenly, without warning, all of a sudden, from nothing एकदम से
17. **Minnow** (noun) – small player, minor entity, underdog, lesser competitor, lightweight छोटा / मामूली पक्ष
18. **Ripple** (noun) – wave, small disturbance, minor effect, small wave, slight movement हलचल
19. **Mere** (adjective) – only, just, simply, no more than, minimal मात्र
20. **Debut** (noun) – first appearance, inaugural showing, introduction, launch, premiere शुरुआत
21. **Nonetheless** (adverb) – nevertheless, however, even so, still, in spite of that इसके बावजूद

22. **Shake** (verb) – unsettle, disturb, rattle, jolt, shock झकझोर देना
23. **Tenet** (noun) – principle, doctrine, belief, core idea, dogma सिद्धांत
24. **Cutting-edge** (adjective) – innovative, pioneering, advanced, leading, state-of-the-art अत्याधुनिक
25. **Hinder** (verb) – hamper, obstruct, impede, hold back, slow down बाधा डालना
26. **Small yard, high fence approach** (noun) – A restrictive or protectionist strategy aiming to guard a small but critical area with high barriers to external competition. सीमित लेकिन संरक्षित क्षेत्र बनाने की रणनीति
27. **Nimble** (adjective) – more agile, quicker, more flexible, more adaptable, faster-moving अधिक फुर्तीला / तेजी से बदलने वाला
28. **Heel-snapping** (adjective) – close-chasing, following closely, nipping at heels, intensely competitive, right behind कड़ी टक्कर देने वाला
29. **As opposed to** (phrase) – instead of, in contrast to, rather than, compared with, different from इसके विपरीत
30. **Proprietary** (adjective) – owned, privately developed, exclusive, patented, restricted स्वामित्व संबंधी
31. **Hint** (at) (verb) – suggest, imply, insinuate, indicate, allude संकेत देना
32. **Indeed** (adverb) – truly, certainly, surely, definitely, undoubtedly वास्तव में
33. **Nurse** (verb) – cultivate, foster, nurture, develop, harbour संजोना
34. **Ambition** (noun) – aspiration, goal, objective, desire, aim महत्वाकांक्षा
35. **Gravitate** (verb) – move towards, be drawn, incline, pull, lean आकर्षित होना
36. **Sovereign** (adjective) – independent, self-governing, autonomous, supreme, having complete authority संप्रभु
37. **Put one's weight** (phrase) – fully support, throw backing behind, lend assistance, commit resources, strongly endorse पूरा समर्थन देना
38. **Commit** (verb) – pledge, dedicate, promise, devote, allocate प्रतिबद्ध होना

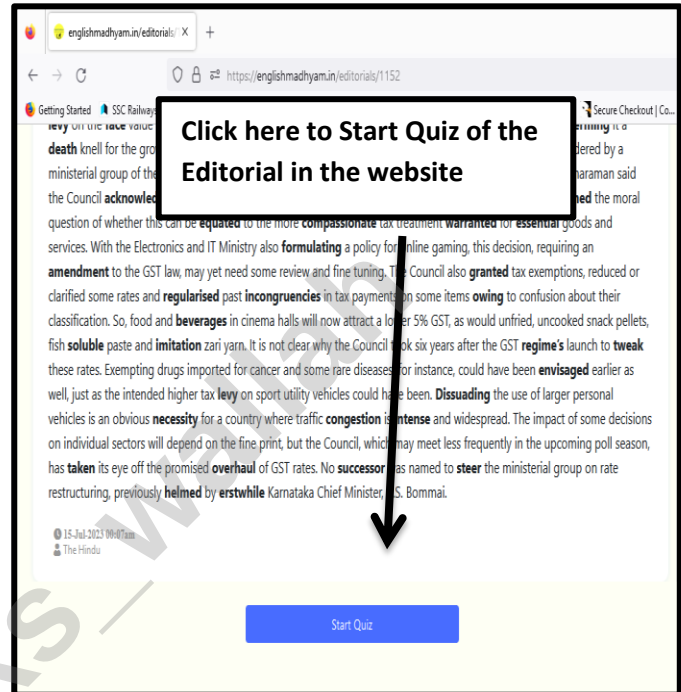
Summary of the Editorial

1. **AI as a Global Battlefield** – The AI race is intensifying, with China's DeepSeek R1 posing a serious challenge to US dominance.
2. **Surprise Emergence of DeepSeek** – Despite being a relatively unknown player, DeepSeek's LLM outperformed Silicon Valley models on several benchmarks.
3. **Wall Street Shock** – US stocks, including Nvidia, saw a massive drop in value, highlighting market concerns over China's AI advancements.
4. **US Chip Export Ban** – The 2022 ban on advanced chip exports to China aimed to maintain America's AI edge but did not prevent Chinese innovation.
5. **DeepSeek's Budget Efficiency** – DeepSeek developed competitive AI models on a budget of \$6 million, compared to OpenAI's estimated \$500 million for ChatGPT.
6. **Breakthrough of R1 Model** – DeepSeek R1, despite using second-tier chips, rivaled OpenAI's latest models, challenging the belief that massive funding is essential for AI progress.
7. **Limitations of US Strategy** – America's "small yard, high fence" approach, restricting access to high-end chips, failed to stop China's AI growth.
8. **Rise of Open-Source AI in China** – Unlike US companies that favor proprietary models, China's open-source AI approach is attracting global talent.
9. **Meta as an Exception** – Among US tech giants, Meta stands out for embracing open-source AI, unlike OpenAI and Google.
10. **China's Influence on AI Rules** – By promoting open-source AI, China is reshaping global AI development and policy frameworks.
11. **Lessons for India** – India can take inspiration from DeepSeek, proving that resource-efficient AI development is possible.
12. **India's AI Ambitions** – The country's **IndiaAI mission**, with a ₹10,300 crore commitment, aims to develop a "sovereign AI" ecosystem.
13. **Need for AI Hardware Access** – To compete globally, India must secure access to advanced AI hardware, including top-tier chips.
14. **Balancing Innovation and Resources** – While budget constraints exist, strategic investments and open-source models can drive AI progress in India.
15. **Ensuring India's AI Future** – If AI defines the future, India must actively participate in its development to remain competitive on the global stage.

Practice Exercise: SSC Pattern Based



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